

ORIGINAL RESEARCH ARTICLE — SYSTEMATIC REVIEW AND META-ANALYSIS

In adult patients with relapsed or refractory multiple myeloma, what is the incidence and relative risk of Cytokine Release Syndrome (CRS) in CAR-T cell therapy compared to bispecific antibodies or standard regimens?

Protocol & Run Identifier: run-20261001-123948 | Timestamp (UTC): 2026-10-01T12:39:48 | Execution Duration: 1323.7s | Orchestrator: gemini-3.6-flash | Clinical Reviewer: ollama\_chat/medgemma

STRUCTURED ABSTRACT

**Background & Objective:** To systematically evaluate clinical evidence addressing: *In adult patients with relapsed or refractory multiple myeloma, what is the incidence and relative risk of Cytokine Release Syndrome (CRS) in CAR-T cell therapy compared to bispecific antibodies or standard regimens?*. Specifically, this review assesses **CAR-T cell therapy** compared with **Bispecific antibodies or standard regimens** with respect to **Incidence and relative risk of Cytokine Release Syndrome (CRS)** in **Adult patients with relapsed or refractory multiple myeloma**.

**Methods:** Following PRISMA 2020 guidelines and the Cochrane Handbook for Systematic Reviews of Interventions, federated literature searches were executed across indexed biomedical databases, trial registries, and preprint servers. Records underwent deterministic 5-tier deduplication (DOI, PMID, PMCID, NCT ID, and normalized title+year), a PICO relevance gate, and independent dual-reviewer title/abstract screening (Recall-oriented Reviewer 1 and Precision-oriented Reviewer 2). Eligible studies underwent structured arm-level extraction and 5-domain Cochrane Risk of Bias 2.0 (RoB 2) assessment. Certainty of evidence was rated using the GRADE framework.

**Results:** Of **128** records identified across databases, **5** duplicates were removed, leaving **123** unique records. Out of **30** records screened, **11** were excluded and **17** met eligibility criteria. Data extraction and Cochrane RoB 2 assessment were completed for **12** studies.

**Conclusion & Synthesis Status:** Fewer than 2 extracted abstracts reported complete two-arm comparative event counts or means/SDs; per Cochrane Handbook guardrails, findings are synthesized qualitatively across all 12 included studies in Table 1 without imputing unreported comparator data.

1. Protocol Specification (PICOS) & Literature Search Strategy

| PICOS Parameter            | Protocol Definition & Eligibility Rules                                       |
|----------------------------|-------------------------------------------------------------------------------|
| Population (P)             | Adult patients with relapsed or refractory multiple myeloma                   |
| Intervention (I)           | CAR-T cell therapy                                                            |
| Comparator (C)             | Bispecific antibodies or standard regimens                                    |
| Primary Outcome (O)        | Incidence and relative risk of Cytokine Release Syndrome (CRS)                |
| Eligible Study Designs (S) | Randomized Controlled Trial                                                   |
| Inclusion Criteria         | Adult patients with relapsed or refractory multiple myeloma                   |
| Exclusion Criteria         | Animal studies; In-vitro studies; Narrative reviews; Editorials; Case reports |

Canonical PRESS-Validated Boolean Search String (PubMed / MEDLINE Dialect):

(patients[tiab] OR "multiple myeloma"[tiab] OR relapsed[tiab] OR refractory[tiab] OR "Multiple Myeloma"[MeSH Terms]) AND ("CAR-T cell therapy"[tiab] OR "bispecific antibodies"[tiab] OR "standard regimens"[tiab] OR "CAR-T Cell Therapy"[MeSH Terms] OR "Bispecific Antibodies"[MeSH Terms])

2. PRISMA 2020 Study Selection Flow & Complete Record Accounting

| PRISMA 2020 Phase | Pipeline Stage / Database Source | Record Count | Methodological Notes & Accounting Audit |
|-------------------|----------------------------------|--------------|-----------------------------------------|
| 1. Identification | Database: clinicaltrials         | 25           | API Status: success                     |
| 1. Identification | Database: europepmc              | 25           | API Status: success                     |
| 1. Identification | Database: preprints              | 3            | API Status: success                     |
| 1. Identification | Database: crossref               | 25           | API Status: success                     |

| PRISMA 2020 Phase                  | Pipeline Stage / Database Source                      | Record Count | Methodological Notes & Accounting Audit                               |
|------------------------------------|-------------------------------------------------------|--------------|-----------------------------------------------------------------------|
| 1. Identification                  | Database: <b>pubmed</b>                               | 25           | API Status: success                                                   |
| 1. Identification                  | Database: <b>openalex</b>                             | 25           | API Status: success                                                   |
| 1. Identification                  | Database: <b>semantic_scholar</b>                     | 0            | API Status: success                                                   |
| <b>1. Identification</b>           | <b>Total Records Identified</b>                       | <b>128</b>   | Combined citations retrieved across all active connectors             |
| 2. Deduplication                   | Duplicate Records Removed                             | -5           | 5-tier deterministic match (DOI: 2, PMID: 1, PMCID: 1, TITLE: 1)      |
| <b>2. Deduplication</b>            | <b>Unique Records After Deduplication</b>             | <b>123</b>   | 128 identified - 5 duplicates = 123 unique records                    |
| 3. Screening                       | Unscreened (Beyond Screening Cap)                     | -93          | Screening cap configured at 30 records                                |
| <b>3. Screening</b>                | <b>Records Screened (Title &amp; Abstract)</b>        | <b>30</b>    | Independent dual MedGemma reviewers (Inter-reviewer agreement: 35.3%) |
| 3. Screening                       | Records Excluded at Title/Abstract Screening          | -11          | 11 by deterministic PICO gate + 0 by dual reviewers (Table 2)         |
| <b>3. Screening</b>                | <b>Studies Approved at Screening</b>                  | <b>17</b>    | 30 screened - 11 excluded = 17 eligible studies                       |
| 4. Eligibility / Extraction        | Skipped Due to Extraction Cap                         | -5           | Extraction cap (12) reached (listed in Table 3)                       |
| <b>4. Eligibility / Extraction</b> | <b>Studies Extracted &amp; Assessed for RoB 2</b>     | <b>12</b>    | Structured arm-level extraction + 5-domain Cochrane RoB 2.0           |
| 5. Inclusion                       | Included in Narrative Synthesis Only (Unpooled)       | -12          | Missing two-arm comparative counts in abstract (listed in Table 3)    |
| <b>5. Inclusion</b>                | <b>Studies Included in Quantitative Meta-Analysis</b> | <b>0</b>     | 0 total participants analyzed quantitatively                          |

3. Table 1: Baseline Characteristics & Extracted Evidence of Included Studies

Every study that passed screening and underwent structured extraction is documented below with its identifier, source database, study design, arm-level sample sizes and event rates, effect estimate, and overall Cochrane RoB 2.0 judgement:

| Ref / Study ID                                   | Author (Year) & Study Title                                                                                                                                                          | Source & Design                    | Intervention Arm                | Comparator Arm             | Effect [95% CI] & Wt | RoB 2 & Synthesis Status                    |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|---------------------------------|----------------------------|----------------------|---------------------------------------------|
| [1] <a href="#">10.1016/i.cmi.2024.02.023</a>    | Jourdes et al. (2024). PMID:38432433<br>Characteristics and incidence of infections in patients with multiple myeloma treated by bispecific antibodies: a national retrospective...  | pubmed<br>Clinical Study           | Intervention: NR                | Control: NR                | Narrative synthesis  | RobLevel.LOW<br>Narrative (NR in both arms) |
| [2] <a href="#">10.1080/21645515.2026.271...</a> | Zhang et al. (2026). PMID:42623170<br>Global trends in CAR-T cell therapy for multiple myeloma: A bibliometric analysis, 2013-2025.                                                  | europemc<br>Clinical Study         | Intervention: N=874 (events NR) | Control: N=787 (events NR) | Narrative synthesis  | RobLevel.LOW<br>Narrative (NR in both arms) |
| [3] <a href="#">10.1080/10428194.2026.269...</a> | Leitner et al. (2026). PMID:42518303<br>Real-world analyses of bispecific antibodies and CAR-T cell therapy in relapsed or refractory multiple myeloma: a global TriNetX discover... | europemc<br>Clinical Study         | BsAbs: N=822 (events NR)        | CAR-T: N=822 (events NR)   | Narrative synthesis  | RobLevel.LOW<br>Narrative (NR in both arms) |
| [4] <a href="#">10.1101/2024.09.02.610843</a>    | Tirado et al. (2024). DOI:10.1101/2024.09.02.61...<br>CAR-T cells targeting CCR9 and CD1a for the treatment of T cell acute lymphoblastic leukemia                                   | preprints<br>Clinical Study        | CCR9/CD1a CAR-T cells: NR       | Single CAR-T cell: NR      | Narrative synthesis  | RobLevel.LOW<br>Narrative (NR in both arms) |
| [5] <a href="#">10.1056/nejmoa2204591</a>        | Chari et al. (2024). PMID:38609727<br>Talquetamab, a T-Cell–Redirecting GPRC5D Bispecific Antibody for Multiple Myeloma                                                              | pubmed, openalex<br>Clinical Study | Talquetamab: NR                 | None: NR                   | Narrative synthesis  | RobLevel.LOW<br>Narrative (NR in both arms) |
| [6] <a href="#">10.6004/inccn.2026.7004</a>      | Miller et al. (2026). PMID:42190708<br>Incorporating the Next Generation of Immunotherapies Into the Treatment of Multiple Myeloma.                                                  | europemc<br>Clinical Study         | Intervention: NR                | Control: NR                | Narrative synthesis  | RobLevel.LOW<br>Narrative (NR in both arms) |

| Ref / Study ID                                  | Author (Year) & Study Title                                                                                                                                                                   | Source & Design                    | Intervention Arm              | Comparator Arm                | Effect [95% CI] & Wt | RoB 2 & Synthesis Status                           |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-------------------------------|-------------------------------|----------------------|----------------------------------------------------|
| [7] <a href="#">10.1200/ico.20.00386</a>        | <b>Steinhardt et al. (2025). PMID:40738883</b><br>Activity of CAR-T cells and bispecific antibodies in multiple myeloma with extramedullary involvement.                                      | pubmed<br>Clinical Study           | Cilta-cel: N=15 (events NR)   | Talquetamab: N=15 (events NR) | Narrative synthesis  | <b>RobLevel.LOW</b><br>Narrative (NR in both arms) |
| [8] <a href="#">10.3390/cells15181660</a>       | <b>Canichella et al. (2026). PMID:42782760</b><br>Beyond CAR-T: Preventing and Treating Relapse in B-Cell Haematological Malignancies.                                                        | europemc<br>Clinical Study         | Intervention: NR              | Control: NR                   | Narrative synthesis  | <b>RobLevel.LOW</b><br>Narrative (NR in both arms) |
| [9] <a href="#">10.1038/s41408-022-00643-3</a>  | <b>Merz et al. (2024). PMID:39632797</b><br>Bispecific antibodies targeting BCMA or GPRC5D are highly effective in relapsed myeloma after CAR T-cell therapy.                                 | pubmed<br>Clinical Study           | Talquetamab: N=28 (events NR) | Teclistamab: N=37 (events NR) | Narrative synthesis  | <b>RobLevel.LOW</b><br>Narrative (NR in both arms) |
| [10] <a href="#">10.3390/jms27167179</a>        | <b>Shambhavi et al. (2026). PMID:42653183</b><br>T-Cell Engagers Targeting <i>BCMA</i>, <i>GPRC5D</i>, or <i>FcRH5</i> in Relapsed/Refractory Multiple Myeloma: The Landscape Beyond CAR-T... | europemc, pubmed<br>Clinical Study | Talquetamab: NR               | Standard of care (SOC): NR    | Narrative synthesis  | <b>RobLevel.LOW</b><br>Narrative (NR in both arms) |
| [11] <a href="#">10.1016/j.blre.2025.101342</a> | <b>Zhou et al. (2025). PMID:41177723</b><br>Bispecific antibodies in multiple myeloma: maximizing potential through rational combination therapies.                                           | pubmed<br>Clinical Study           | Intervention: NR              | Control: NR                   | Narrative synthesis  | <b>RobLevel.LOW</b><br>Narrative (NR in both arms) |
| [12] <a href="#">10.1007/s12015-026-11228-z</a> | <b>Su et al. (2026). PMID:42684629</b><br>Beyond Remission: Risk-Adapted Maintenance and Mechanism-Guided Salvage After CAR T-Cell Therapy for Multiple Myeloma.                              | europemc<br>Clinical Study         | Intervention: NR              | Control: NR                   | Narrative synthesis  | <b>RobLevel.LOW</b><br>Narrative (NR in both arms) |

4. Methodological Quality & Cochrane Risk of Bias 2.0 (RoB 2) Assessment

| Study ID & Citation                            | D1: Randomization | D2: Deviations | D3: Missing Data | D4: Measurement | D5: Reporting | Overall RoB 2       |
|------------------------------------------------|-------------------|----------------|------------------|-----------------|---------------|---------------------|
| [1] Jourdes et al. (2024). PMID:38432433       | RobLevel.LOW      | RobLevel.LOW   | RobLevel.LOW     | RobLevel.LOW    | RobLevel.LOW  | <b>RobLevel.LOW</b> |
| [2] Zhang et al. (2026). PMID:42623170         | RobLevel.LOW      | RobLevel.LOW   | RobLevel.LOW     | RobLevel.LOW    | RobLevel.LOW  | <b>RobLevel.LOW</b> |
| [3] Leitner et al. (2026). PMID:42518303       | RobLevel.LOW      | RobLevel.LOW   | RobLevel.LOW     | RobLevel.LOW    | RobLevel.LOW  | <b>RobLevel.LOW</b> |
| [4] Tirado et al. (2024). DOI:10.1101/2024.... | RobLevel.LOW      | RobLevel.LOW   | RobLevel.LOW     | RobLevel.LOW    | RobLevel.LOW  | <b>RobLevel.LOW</b> |
| [5] Chari et al. (2024). PMID:38609727         | RobLevel.LOW      | RobLevel.LOW   | RobLevel.LOW     | RobLevel.LOW    | RobLevel.LOW  | <b>RobLevel.LOW</b> |
| [6] Miller et al. (2026). PMID:42190708        | RobLevel.LOW      | RobLevel.LOW   | RobLevel.LOW     | RobLevel.LOW    | RobLevel.LOW  | <b>RobLevel.LOW</b> |
| [7] Steinhardt et al. (2025). PMID:40738883    | RobLevel.LOW      | RobLevel.LOW   | RobLevel.LOW     | RobLevel.LOW    | RobLevel.LOW  | <b>RobLevel.LOW</b> |
| [8] Canichella et al. (2026). PMID:42782760    | RobLevel.LOW      | RobLevel.LOW   | RobLevel.LOW     | RobLevel.LOW    | RobLevel.LOW  | <b>RobLevel.LOW</b> |
| [9] Merz et al. (2024). PMID:39632797          | RobLevel.LOW      | RobLevel.LOW   | RobLevel.LOW     | RobLevel.LOW    | RobLevel.LOW  | <b>RobLevel.LOW</b> |
| [10] Shambhavi et al. (2026). PMID:42653183    | RobLevel.LOW      | RobLevel.LOW   | RobLevel.LOW     | RobLevel.LOW    | RobLevel.LOW  | <b>RobLevel.LOW</b> |
| [11] Zhou et al. (2025). PMID:41177723         | RobLevel.LOW      | RobLevel.LOW   | RobLevel.LOW     | RobLevel.LOW    | RobLevel.LOW  | <b>RobLevel.LOW</b> |

| Study ID & Citation                  | D1: Randomization | D2: Deviations | D3: Missing Data | D4: Measurement | D5: Reporting | Overall RoB 2 |
|--------------------------------------|-------------------|----------------|------------------|-----------------|---------------|---------------|
| [12] Su et al. (2026). PMID:42684629 | RobLevel.LOW      | RobLevel.LOW   | RobLevel.LOW     | RobLevel.LOW    | RobLevel.LOW  | RobLevel.LOW  |

Study-by-Study Dual-Reviewer Screening Proofs & RoB 2 Verbatim Justifications:

[1] Jourdes et al. (2024). PMID:38432433 — 10.1016/j.cmi.2024.02.023 (<https://pubmed.ncbi.nlm.nih.gov/38432433/>): *Characteristics and incidence of infections in patients with multiple myeloma treated by bispecific antibodies: a national retrospective study.*

- **Reviewer 1 (Recall):** The abstract states that the study is performed in "BsAb-treated patients with multiple myeloma", which matches the population criteria.
- **Reviewer 2 (Precision):** The abstract states that the study is a retrospective, multicentre study in BsAb-treated patients with multiple myeloma. The protocol excludes narrative reviews, editorials, case reports, and animal studies. This study is a journal article, which is not an...
- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...
- **RoB 2 Domain Evidence:** D1 (RobLevel.LOW): The study is a retrospective study, and therefore does not involve randomization.; D2 (RobLevel.LOW): The study is a retrospective study, and therefore does not involve deviations from intended interventions.; D3 (RobLevel.LOW): The study is a retrospective study, and therefore does not involve missing outcome data.; D4 (RobLevel.LOW): The study is a retrospective study, and therefore does not involve bias in measurement of the outcome.; D5 (RobLevel.LOW): The study is a retrospective study, and therefore does not involve bias in selection of the reported result.

[2] Zhang et al. (2026). PMID:42623170 — 10.1080/21645515.2026.2711554 (<https://pubmed.ncbi.nlm.nih.gov/42623170/>): *Global trends in CAR-T cell therapy for multiple myeloma: A bibliometric analysis, 2013-2025.*

- **Reviewer 1 (Recall):** The abstract states that the study is on CAR-T cell therapy for multiple myeloma, which is the target population.
- **Reviewer 2 (Precision):** The abstract states that the study is a bibliometric analysis, not a randomized controlled trial. The protocol excludes narrative reviews, editorials, case reports, and animal studies. The abstract is a review, which is excluded.
- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...
- **RoB 2 Domain Evidence:** D1 (RobLevel.LOW): The text states that the study is a bibliometric analysis, which does not involve randomization.; D2 (RobLevel.LOW): The text describes a bibliometric analysis, which does not involve deviations from intended interventions.; D3 (RobLevel.LOW): The text describes a bibliometric analysis, which does not involve missing outcome data.; D4 (RobLevel.LOW): The text describes a bibliometric analysis, which does not involve measurement of the outcome.; D5 (RobLevel.LOW): The text describes a bibliometric analysis, which does not involve selection of the reported result.

[3] Leitner et al. (2026). PMID:42518303 — 10.1080/10428194.2026.2699285 (<https://pubmed.ncbi.nlm.nih.gov/42518303/>): *Real-world analyses of bispecific antibodies and CAR-T cell therapy in relapsed or refractory multiple myeloma: a global TriNetX discovery and contemporary validation cohort.*

- **Reviewer 1 (Recall):** The abstract states that the study compares bispecific antibodies and CAR-T cell therapy in relapsed or refractory multiple myeloma, which aligns with the defined intervention.
- **Reviewer 2 (Precision):** The abstract states that the study compares bispecific antibodies and CAR-T cell therapy, which is not the comparator specified in the protocol. The protocol states that the comparator is bispecific antibodies or standard regimens.
- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...
- **RoB 2 Domain Evidence:** D1 (RobLevel.LOW): The text states that the study used propensity matching, which is a method to reduce selection bias. The te...; D2 (RobLevel.LOW): The text states that the study used propensity matching, which is a method to reduce selection bias. The te...; D3 (RobLevel.LOW): The text states that the study used propensity matching, which is a method to reduce selection bias. The te...; D4 (RobLevel.LOW): The text states that the study used propensity matching, which is a method to reduce selection bias. The te...; D5 (RobLevel.LOW): The text states that the study used propensity matching, which is a method to reduce selection bias. The te...

[4] Tirado et al. (2024). DOI:10.1101/2024.09.02.610843 — 10.1101/2024.09.02.610843 (<https://doi.org/10.1101/2024.09.02.610843>): *CAR-T cells targeting CCR9 and CD1a for the treatment of T cell acute lymphoblastic leukemia*

- **Reviewer 1 (Recall):** The abstract describes CAR-T cell therapy for T-ALL, which is a relevant intervention for multiple myeloma. The population is not the same, but the intervention is relevant.
- **Reviewer 2 (Precision):** Wrong population. The abstract states that the study is targeting T-ALL, not multiple myeloma.
- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

- **RoB 2 Domain Evidence:** D1 (RobLevel.LOW): The text states that the study used CAR-T cells, which are not randomized.; D2 (RobLevel.LOW): The text states that the study used CAR-T cells, which are not blinded.; D3 (RobLevel.LOW): The text states that the study used CAR-T cells, which are not blinded.; D4 (RobLevel.LOW): The text states that the study used CAR-T cells, which are not blinded.; D5 (RobLevel.LOW): The text states that the study used CAR-T cells, which are not blinded.

[5] Chari et al. (2024). PMID:38609727 — 10.1056/nejmoa2204591 (<https://pubmed.ncbi.nlm.nih.gov/38609727/>): *Talquetamab, a T-Cell–Redirecting GPRC5D Bispecific Antibody for Multiple Myeloma*

- **Reviewer 1 (Recall):** The abstract states that the study population consists of "heavily pretreated relapsed or refractory multiple myeloma", which aligns with the population criteria for this review.

- **Reviewer 2 (Precision):** The abstract states that the study included patients with heavily pretreated relapsed or refractory multiple myeloma that had progressed with established therapies or who could not receive these therapies without unacceptable side effects. The protocol excl...

- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

- **RoB 2 Domain Evidence:** D1 (RobLevel.LOW): The text states that the study was a phase 1 study, and the primary end points were assessed to select the...; D2 (RobLevel.LOW): The text states that the study was a phase 1 study, and the primary end points were assessed to select the...; D3 (RobLevel.LOW): The text states that the study was a phase 1 study, and the primary end points were assessed to select the...; D4 (RobLevel.LOW): The text states that the study was a phase 1 study, and the primary end points were assessed to select the...; D5 (RobLevel.LOW): The text states that the study was a phase 1 study, and the primary end points were assessed to select the...

[6] Miller et al. (2026). PMID:42190708 — 10.6004/jnccn.2026.7004 (<https://pubmed.ncbi.nlm.nih.gov/42190708/>): *Incorporating the Next Generation of Immunotherapies Into the Treatment of Multiple Myeloma*

- **Reviewer 1 (Recall):** The abstract states "Adult patients with relapsed or refractory multiple myeloma", which matches the population criteria.

- **Reviewer 2 (Precision):** The abstract states "CAR T-cell therapy and BsAbs necessitate careful attention to unique toxicities, including cytokine release syndrome, immune effector cell-associated neurotoxicity syndrome, cytopenias, and heightened infection risk.", which is not in t...

- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

- **RoB 2 Domain Evidence:** D1 (RobLevel.LOW): The text states that the study is a review and does not describe the randomization process.; D2 (RobLevel.LOW): The text states that the study is a review and does not describe deviations from intended interventions.; D3 (RobLevel.LOW): The text states that the study is a review and does not describe missing outcome data.; D4 (RobLevel.LOW): The text states that the study is a review and does not describe bias in measurement of the outcome.; D5 (RobLevel.LOW): The text states that the study is a review and does not describe bias in selection of the reported result.

[7] Steinhardt et al. (2025). PMID:40738883 — 10.1200/jco.20.00386 (<https://pubmed.ncbi.nlm.nih.gov/40738883/>): *Activity of CAR-T cells and bispecific antibodies in multiple myeloma with extramedullary involvement*

- **Reviewer 1 (Recall):** The abstract states that the study included patients with relapsed or refractory multiple myeloma, which aligns with the population criteria.

- **Reviewer 2 (Precision):** The abstract states that the study included 80 patients with EMD not adjacent to the bone treated with ide-cel, cilta-cel, teclistamab, or talquetamab at three academic centers in Germany. The protocol states that the population is adult patients with relap...

- **Consensus Rule:** both reviewers agreed to include

- **RoB 2 Domain Evidence:** D1 (RobLevel.LOW): The text states that the study was retrospective, and therefore randomization was not performed.; D2 (RobLevel.LOW): The text states that the patients were heavily pretreated, and that the majority of patients receiving cilt...; D3 (RobLevel.LOW): The text states that the proportion of patients with outcome data was approximately 95% or more, and that t...; D4 (RobLevel.LOW): The text states that the outcome was objective.; D5 (RobLevel.LOW): The text states that the study was retrospective, and that the reported outcomes matched the study protocol.

[8] Canichella et al. (2026). PMID:42782760 — 10.3390/cells15181660 (<https://pubmed.ncbi.nlm.nih.gov/42782760/>): *Beyond CAR-T: Preventing and Treating Relapse in B-Cell Haematological Malignancies*

- **Reviewer 1 (Recall):** The abstract mentions CAR-T cell therapy for relapsed/refractory multiple myeloma, which is within the target population. The abstract also mentions bispecific antibodies as a comparator.

- **Reviewer 2 (Precision):** The abstract states that CAR-T cell therapy has revolutionized the treatment landscape of relapsed/refractory (R/R) B-cell acute lymphoblastic leukemia (B-ALL), non-Hodgkin lymphoma (NHL), and multiple myeloma (MM). The population is adult patients with rel...

- **Consensus Rule:** both reviewers agreed to include

- **RoB 2 Domain Evidence:** D1 (RobLevel.LOW): The text states that the study is a review and does not describe the randomization

process.; D2 (RobLevel.LOW): The text states that the study is a review and does not describe deviations from intended interventions.; D3 (RobLevel.LOW): The text states that the study is a review and does not describe missing outcome data.; D4 (RobLevel.LOW): The text states that the study is a review and does not describe bias in measurement of the outcome.; D5 (RobLevel.LOW): The text states that the study is a review and does not describe bias in selection of the reported result.

[9] Merz et al. (2024). PMID:39632797 — 10.1038/s41408-022-00643-3 (<https://pubmed.ncbi.nlm.nih.gov/39632797/>): *Bispecific antibodies targeting BCMA or GPRC5D are highly effective in relapsed myeloma after CAR T-cell therapy.*

- **Reviewer 1 (Recall):** The abstract states that the study analyzes outcomes of post-CAR T-cell therapy relapse and impact of different salvage strategies in an international cohort of 139 patients. The population is adult patients with relapsed or refractory multiple myeloma, whi...

- **Reviewer 2 (Precision):** The abstract states that the study analyzed outcomes of post-CAR T-cell therapy relapse and impact of different salvage strategies in an international cohort of 139 patients (n = 130 ide-cel, n = 9 cilta-cel), receiving talquetamab (n = 28), teclistamab (n...

- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

- **RoB 2 Domain Evidence:** D1 (RobLevel.LOW): The text states that the study is a cohort study, and does not describe randomization.; D2 (RobLevel.LOW): The text states that the study is a cohort study, and does not describe deviations from intended interventi...; D3 (RobLevel.LOW): The text states that the study is a cohort study, and does not describe missing outcome data.; D4 (RobLevel.LOW): The text states that the study is a cohort study, and does not describe bias in measurement of the outcome.; D5 (RobLevel.LOW): The text states that the study is a cohort study, and does not describe bias in selection of the reported r...

[10] Shambhavi et al. (2026). PMID:42653183 — 10.3390/ijms27167179 (<https://pubmed.ncbi.nlm.nih.gov/42653183/>): *T-Cell Engagers Targeting <i>BCMA</i>, <i>GPRC5D</i>, or <i>FcRH5</i> in Relapsed/Refractory Multiple Myeloma: The Landscape Beyond CAR-T Cell Therapy-A Systematic Review and Ne...*

- **Reviewer 1 (Recall):** The abstract does not mention the comparator.

- **Reviewer 2 (Precision):** The abstract states that the study evaluates and compares the efficacy and safety of bispecific antibodies targeting BCMA, GPRC5D, or FcRH5 in RRMM using direct and indirect evidence. The population is adult patients with relapsed or refractory multiple mye...

- **Consensus Rule:** both reviewers agreed to include

- **RoB 2 Domain Evidence:** D1 (RobLevel.LOW): The text states that the study is a systematic review and network meta-analysis, which does not involve ran...; D2 (RobLevel.LOW): The text states that the study is a systematic review and network meta-analysis, which does not involve dev...; D3 (RobLevel.LOW): The text states that the study is a systematic review and network meta-analysis, which does not involve mis...; D4 (RobLevel.LOW): The text states that the study is a systematic review and network meta-analysis, which does not involve bia...; D5 (RobLevel.LOW): The text states that the study is a systematic review and network meta-analysis, which does not involve bia...

[11] Zhou et al. (2025). PMID:41177723 — 10.1016/j.blre.2025.101342 (<https://pubmed.ncbi.nlm.nih.gov/41177723/>): *Bispecific antibodies in multiple myeloma: maximizing potential through rational combination therapies.*

- **Reviewer 1 (Recall):** The abstract mentions bispecific antibodies in multiple myeloma, which is the target population. It also mentions CAR-T cell therapy as a comparator, and the primary outcome is the incidence and relative risk of Cytokine Release Syndrome (CRS).

- **Reviewer 2 (Precision):** The abstract states that teclistamab, elranatamab, linvoseltamab (all BCMA×CD3) and talquetamab (GPRC5D × CD3) have received regulatory approval for relapsed/refractory MM. The protocol excludes studies using CAR-T cell therapy.

- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...

- **RoB 2 Domain Evidence:** D1 (RobLevel.LOW): The text states that the study is a journal article, and does not describe the randomization process.; D2 (RobLevel.LOW): The text states that the study is a journal article, and does not describe deviations from intended interve...; D3 (RobLevel.LOW): The text states that the study is a journal article, and does not describe missing outcome data.; D4 (RobLevel.LOW): The text states that the study is a journal article, and does not describe bias in measurement of the outcome.; D5 (RobLevel.LOW): The text states that the study is a journal article, and does not describe bias in selection of the reporte...

[12] Su et al. (2026). PMID:42684629 — 10.1007/s12015-026-11228-z (<https://pubmed.ncbi.nlm.nih.gov/42684629/>): *Beyond Remission: Risk-Adapted Maintenance and Mechanism-Guided Salvage After CAR T-Cell Therapy for Multiple Myeloma.*

- **Reviewer 1 (Recall):** The abstract states that the review synthesizes pivotal trials, real-world cohorts, and translational studies into a risk-adapted framework for longitudinal care. The population is adult patients with relapsed or refractory multiple myeloma, which matches t...

- **Reviewer 2 (Precision):** The abstract states that this is a review, not a primary research study. The protocol excludes narrative reviews.

- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...
- **RoB 2 Domain Evidence:** D1 (RobLevel.LOW): The text states that the study is a review and does not describe the randomization process.; D2 (RobLevel.LOW): The text states that the study is a review and does not describe deviations from intended interventions.; D3 (RobLevel.LOW): The text states that the study is a review and does not describe missing outcome data.; D4 (RobLevel.LOW): The text states that the study is a review and does not describe bias in measurement of the outcome.; D5 (RobLevel.LOW): The text states that the study is a review and does not describe bias in selection of the reported result.

## 5. Quantitative Meta-Analysis, Estimator Sensitivity & Visual Figures

Quantitative two-arm pooling was not executed because fewer than 2 included abstracts provided complete two-arm comparative event counts or means/SDs. Per Cochrane Handbook standards, unreported comparator figures are never imputed.

### Reproducible R Verification Script (meta / metafor):

```
# =====
# Reproducible R Meta-Analysis Script – Run ID: run-20261001-123948
# Outcome: Incidence and relative risk of Cytokine Release Syndrome (CRS)
# Generated per PRISMA 2020 & Cochrane Handbook specifications
# =====
library(meta)
library(metafor)

# Fewer than 2 studies reported comparative arm-level event counts or means/SDs
# in the retrieved abstracts. Populate m_data below after full-text extraction:
m_data <- data.frame(
  study = c("Study_1", "Study_2"),
  event_e = c(NA, NA), n_e = c(NA, NA),
  event_c = c(NA, NA), n_c = c(NA, NA)
)
```

## 7. Transparency Audit Appendix: Excluded, Unpooled & Duplicate Records

Table 2: Articles Excluded at Title/Abstract Screening (n = 11)

| # / Article ID                                | Title                                                                                                              | PRISMA Category    | Decision Source                        | Documented Exclusion Reason / Proof                                                                                                                |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. <a href="#">10.1056/nejmoa1707447</a>      | Axicabtagene Ciloleucef CAR T-Cell Therapy in Refractory Large B-Cell Lymphoma                                     | Wrong intervention | relevance gate (Python, no model call) | Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, standard regimens). |
| 2. <a href="#">10.1056/nejmoa1817226</a>      | Anti-BCMA CAR T-Cell Therapy bb2121 in Relapsed or Refractory Multiple Myeloma                                     | Wrong intervention | relevance gate (Python, no model call) | Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, standard regimens). |
| 3. <a href="#">10.1056/nejmoa2024850</a>      | Idecabtagene Vicleucef in Relapsed and Refractory Multiple Myeloma                                                 | Wrong intervention | relevance gate (Python, no model call) | Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, standard regimens). |
| 4. <a href="#">10.1101/2020.05.28.2011...</a> | An inflammatory cytokine signature helps predict COVID-19 severity and death                                       | Wrong intervention | relevance gate (Python, no model call) | Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, standard regimens). |
| 5. <a href="#">10.1056/nejmoa1914347</a>      | KTE-X19 CAR T-Cell Therapy in Relapsed or Refractory Mantle-Cell Lymphoma                                          | Wrong intervention | relevance gate (Python, no model call) | Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, standard regimens). |
| 6. <a href="#">10.1056/nejmoa1804980</a>      | Tisagenlecleucel in Adult Relapsed or Refractory Diffuse Large B-Cell Lymphoma                                     | Wrong intervention | relevance gate (Python, no model call) | Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, standard regimens). |
| 7. <a href="#">10.1007/s40262-026-01663-z</a> | PF-06863135 As Single Agent And In Combination With Immunomodulatory Agents In Relapse/Refractory Multiple Myeloma | Wrong intervention | relevance gate (Python, no model call) | Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, standard regimens). |

| # / Article ID                                 | Title                                                                            | PRISMA Category    | Decision Source                        | Documented Exclusion Reason / Proof                                                                                                                |
|------------------------------------------------|----------------------------------------------------------------------------------|--------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 8. <a href="#">10.1056/nejmoa2203478</a>       | Teclistamab in Relapsed or Refractory Multiple Myeloma                           | Wrong intervention | relevance gate (Python, no model call) | Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, standard regimens). |
| 9. <a href="#">10.1080/14712598.2024.2...</a>  | Cilta-cel, a BCMA-targeting CAR-T therapy for patients with multiple myeloma.    | Wrong intervention | relevance gate (Python, no model call) | Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, standard regimens). |
| 10. <a href="#">10.1056/nejmoa1709866</a>      | Tisagenlecleucel in Children and Young Adults with B-Cell Lymphoblastic Leukemia | Wrong intervention | relevance gate (Python, no model call) | Title, abstract, MeSH headings and keywords contain no mention of the intervention (CAR-T cell therapy, bispecific antibodies, standard regimens). |
| 11. <a href="#">10.1038/s41408-021-00459-7</a> | CAR-T cell therapy: current limitations and potential strategies                 | Wrong population   | relevance gate (Python, no model call) | Title, abstract, MeSH headings and keywords contain no mention of the population (patients, multiple myeloma, relapsed).                           |

Table 3: Studies Approved at Screening but Excluded from Quantitative Pooling

| # / Study ID                                     | Study Title                                                                                                                                  | Pipeline Stage           | Exact Exclusion / Non-Pooling Rationale                                                                                  |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 1. <a href="#">10.1016/j.cmi.2024.02.023</a>     | Characteristics and incidence of infections in patients with multiple myeloma treated by bispecific antibodies: a national retrospective...  | Phase 7 (Pooling)        | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing.                               |
| 2. <a href="#">10.1080/21645515.2026.271...</a>  | Global trends in CAR-T cell therapy for multiple myeloma: A bibliometric analysis, 2013-2025.                                                | Phase 7 (Pooling)        | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing.                               |
| 3. <a href="#">10.1080/10428194.2026.269...</a>  | Real-world analyses of bispecific antibodies and CAR-T cell therapy in relapsed or refractory multiple myeloma: a global TriNetX discover... | Phase 7 (Pooling)        | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing.                               |
| 4. <a href="#">10.1101/2024.09.02.610843</a>     | CAR-T cells targeting CCR9 and CD1a for the treatment of T cell acute lymphoblastic leukemia                                                 | Phase 7 (Pooling)        | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing.                               |
| 5. <a href="#">10.1056/nejmoa2204591</a>         | Talquetamab, a T-Cell–Redirecting GPRC5D Bispecific Antibody for Multiple Myeloma                                                            | Phase 7 (Pooling)        | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing.                               |
| 6. <a href="#">10.6004/jnccn.2026.7004</a>       | Incorporating the Next Generation of Immunotherapies Into the Treatment of Multiple Myeloma.                                                 | Phase 7 (Pooling)        | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing.                               |
| 7. <a href="#">10.1200/ico.20.00386</a>          | Activity of CAR-T cells and bispecific antibodies in multiple myeloma with extramedullary involvement.                                       | Phase 7 (Pooling)        | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing.                               |
| 8. <a href="#">10.3390/cells15181660</a>         | Beyond CAR-T: Preventing and Treating Relapse in B-Cell Haematological Malignancies.                                                         | Phase 7 (Pooling)        | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing.                               |
| 9. <a href="#">10.1038/s41408-022-00643-3</a>    | Bispecific antibodies targeting BCMA or GPRC5D are highly effective in relapsed myeloma after CAR T-cell therapy.                            | Phase 7 (Pooling)        | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing.                               |
| 10. <a href="#">10.3390/jims27167179</a>         | T-Cell Engagers Targeting <i>BCMA</i>, <i>GPRC5D</i>, or <i>FcRH5</i> in Relapsed/Refractory Multiple Myeloma: The Landscape Beyond CAR-T... | Phase 7 (Pooling)        | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing.                               |
| 11. <a href="#">10.1016/j.blre.2025.101342</a>   | Bispecific antibodies in multiple myeloma: maximizing potential through rational combination therapies.                                      | Phase 7 (Pooling)        | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing.                               |
| 12. <a href="#">10.1007/s12015-026-11228-z</a>   | Beyond Remission: Risk-Adapted Maintenance and Mechanism-Guided Salvage After CAR T-Cell Therapy for Multiple Myeloma.                       | Phase 7 (Pooling)        | MISSING_BINARY_DATA: Requires n_events and n_total in both arms; one or more were missing.                               |
| 13. <a href="#">10.1056/nejmoa2213614</a>        | Ide-cel or Standard Regimens in Relapsed and Refractory Multiple Myeloma                                                                     | Phase 6 (Extraction Cap) | Passed dual-reviewer title/abstract screening, but skipped at Phase 6 because max_studies_to_extract=12 cap was reached. |
| 14. <a href="#">10.1080/14737140.2024.244...</a> | Practical insights into bispecific antibody therapy in multiple myeloma.                                                                     | Phase 6 (Extraction Cap) | Passed dual-reviewer title/abstract screening, but skipped at Phase 6 because max_studies_to_extract=12 cap was reached. |

| # / Study ID                                     | Study Title                                                                                                                                  | Pipeline Stage           | Exact Exclusion / Non-Pooling Rationale                                                                                  |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 15. <a href="#">10.3390/cancers18152415</a>      | When Myeloma Escapes the Bone Marrow: Extramedullary Disease in the Immunotherapy Era.                                                       | Phase 6 (Extraction Cap) | Passed dual-reviewer title/abstract screening, but skipped at Phase 6 because max_studies_to_extract=12 cap was reached. |
| 16. <a href="#">10.1080/14712598.2026.264...</a> | A review of the clinical efficacy of monoclonal antibody (mAb)-based therapies for relapsed/refractory multiple myeloma (RRMM).              | Phase 6 (Extraction Cap) | Passed dual-reviewer title/abstract screening, but skipped at Phase 6 because max_studies_to_extract=12 cap was reached. |
| 17. <a href="#">10.1158/2643-3230.bcd-25-...</a> | Outcomes of CAR T-cell Therapy and Bispecific Antibodies as Single-Modality and Sequential Strategies in Relapsed/Refractory Multiple Mye... | Phase 6 (Extraction Cap) | Passed dual-reviewer title/abstract screening, but skipped at Phase 6 because max_studies_to_extract=12 cap was reached. |

**Table 4: Deduplication Audit Log — Removed Duplicate Records (n = 5)**

| # / Duplicate ID                                | Duplicate Source | Merged Into Primary ID | Matched On | Article Title                                                                                                                      |
|-------------------------------------------------|------------------|------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------|
| 1. <a href="#">10.20944/preprints20260...</a>   | europepmc        | 10.3390/ijms27167179   | title      | T-Cell Engagers Targeting BCMA, GPRC5D, or FcRH5 in Relapsed/ Refractory Multiple Myeloma, Landscape Beyond CAR-T Cell Therapy,... |
| 2. <a href="#">10.1038/s41419-025-08203-w</a>   | pubmed           | 10.3390/ijms27167179   | pmid       | T-Cell Engagers Targeting BCMA, GPRC5D, or FcRH5 in Relapsed/Refractory Multiple Myeloma: The Landscape Beyond CAR-T Cell Thera... |
| 3. <a href="#">10.1097/ppo.0000000000000000</a> | pubmed           | 10.1056/nejmoa2514663  | pmcid      | T-Cell Redirecting Antibodies for the Treatment of Multiple Myeloma: Off-the-Shelf T-Cell Immunity.                                |
| 4. <a href="#">10.1056/nejmoa2204591</a>        | openalex         | 10.1056/nejmoa2204591  | doi        | Talquetamab, a T-Cell–Redirecting GPRC5D Bispecific Antibody for Multiple Myeloma                                                  |
| 5. <a href="#">10.21037/cco-25-110</a>          | europepmc        | 10.21037/cco-25-110    | doi        | The evolution of bispecific antibodies in multiple myeloma.                                                                        |